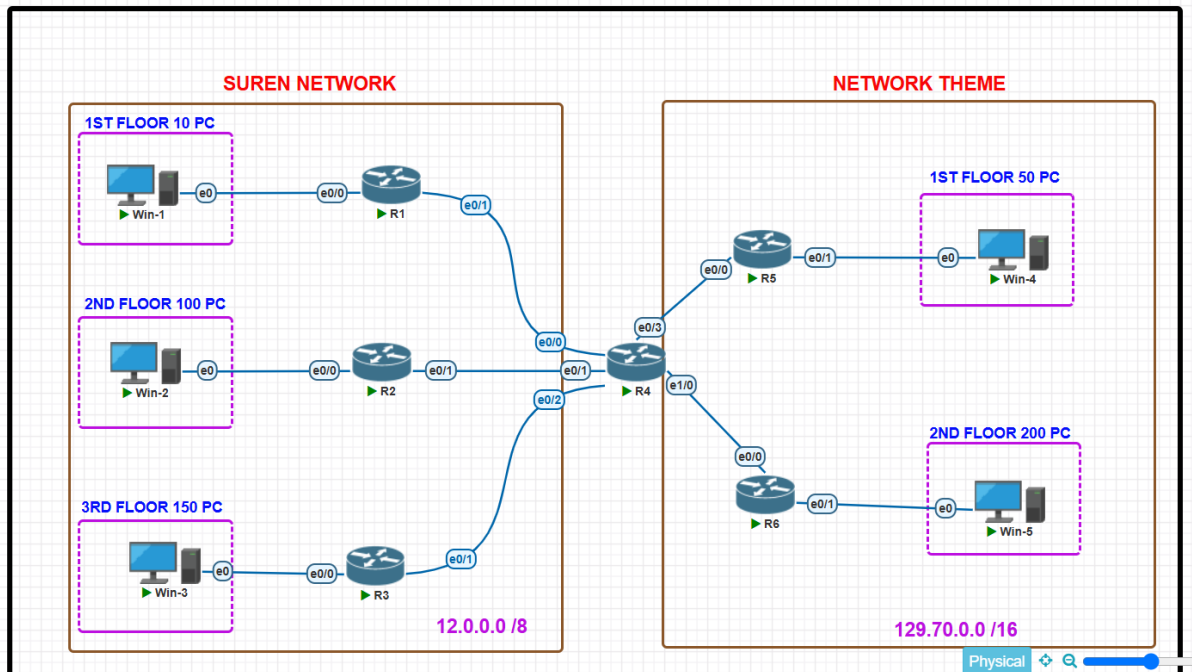


Subnetting



```
R1
R1(config)#interface eth0/1
R1(config-if)#
R1(config-if)#ip add 12.0.1.145 255.255.255.252
R1(config-if)#
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#exit
R1(config)#
R1(config)#
R1(config)#
*Oct 2 15:25:19.893: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#
R1(config)#interface eth0/0
R1(config-if)#
R1(config-if)#
R1(config-if)#ip add 12.0.1.129 255.255.255.240
R1(config-if)#
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#exit
R1(config)#
R1(config)#
R1(config)#
R1(config)#
```

```
R2
R2(config)#interface eth0/1
R2(config-if)#
R2(config-if)#ip add 12.0.1.149 255.255.255.252
R2(config-if)#
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#exit
R2(config)#
R2(config)#
R2(config)#
*Oct  2 15:25:27.422: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#
R2(config)#interface eth0/0
R2(config-if)#
R2(config-if)#ip add 12.0.1.1 255.255.255.128
R2(config-if)#
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#exit
R2(config)#
R2(config)#
R2(config)#
R2(config)#
```

```
R3
R3#
R3#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#
R3(config)#interface eth0/1
R3(config-if)#
R3(config-if)#
R3(config-if)#ip add 12.0.1.153 255.255.255.252
R3(config-if)#
R3(config-if)#no shut
R3(config-if)#
R3(config-if)#
R3(config-if)#exit
R3(config)#
R3(config)#
R3(config)#exit
R3#
*Oct  2 15:26:48.665: %SYS-5-CONFIG_I: Configured from console by console
R3#ping 12.0.1.153
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.1.153, timeout is 2 seconds:
!!!!!!
```

```
R3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.1.153, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/4/5 ms
R3#
*Oct 2 15:27:09.086: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R3#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#
R3(config)#
R3(config)#
R3(config)#interface eth0/0
R3(config-if)#
R3(config-if)#
R3(config-if)#ip add 12.0.0.1 255.255.255.0
R3(config-if)#
R3(config-if)#no shut
R3(config-if)#
R3(config-if)#
R3(config-if)#exit
R3(config)#
R3(config)#
R3(config)#
```

```
R4
Router(config)#
Router(config)#
Router(config)#interface eth0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#ip add 12.0.1.146 255.255.255.252
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#ex
*Oct 2 15:34:40.646: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Oct 2 15:34:41.648: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
*Oct 2 15:34:47.804: %SYS-5-CONFIG_I: Configured from console by console
Router#ping 12.0.1.145
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.1.145, timeout is 2 seconds:
!!!!
```

```
R4
Router(config)#interface eth0/1
Router(config-if)#
Router(config-if)#
Router(config-if)#ip add 12.0.1.150 255.255.255.252
Router(config-if)#
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
*Oct 2 15:36:23.626: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Oct 2 15:36:24.626: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
Router(config)#
Router(config)#
Router(config)#interface eth0/2
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#ip add 12.0.1.154 255.255.255.252
Router(config-if)#
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#exit
```

```
R4
Router(config)#
Router(config)#
Router(config)#interface eth0/3
Router(config-if)#
Router(config-if)#ip add 129.70.1.65 255.255.255.252
Router(config-if)#
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#exit
Router(config)#
*Oct 2 15:38:00.103: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to up
*Oct 2 15:38:01.106: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/3, changed state to up
Router(config)#
Router(config)#interface eth1/0
Router(config-if)#
Router(config-if)#ip add 129.70.1.69 255.255.255.252
Router(config-if)#
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#exit
Router(config)#
*Oct 2 15:38:54.693: %LINK-3-UPDOWN: Interface Ethernet1/0, changed state to up
*Oct 2 15:38:55.693: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed state to up
```

```
R4
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#ip route 12.0.0.0 255.255.255.0 12.0.1.153
Router(config)#
Router(config)#
Router(config)#ip route 12.0.1.0 255.255.255.128 12.0.1.150
%Invalid next hop address (it's this router)
Router(config)#ip route 12.0.1.0 255.255.255.128 12.0.1.149
Router(config)#
Router(config)#
Router(config)#ip route 12.0.1.128 255.255.255.240 12.0.1.145
Router(config)#
Router(config)#exit
Router#
*Oct  2 15:44:16.865: %SYS-5-CONFIG_I: Configured from console by console
Router#
Router#ping 12.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
Router#
```

```
R4
Router>
Router>enable
Router#
Router#
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#ip route 129.70.0.0 255.255.255.0 129.70.1.70
Router(config)#
Router(config)#ip route 129.70.1.0 255.255.255.192 129.70.1.66
Router(config)#
Router(config)#
Router(config)#exit
Router#
*Oct  2 16:07:24.452: %SYS-5-CONFIG_I: Configured from console by console
Router#ping 129.70.1.66
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 129.70.1.66, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router#
```

```
R5
R5(config)#
R5(config)#interface eth0/0
R5(config-if)#
R5(config-if)#
R5(config-if)#ip add 129.70.1.66 255.255.255.252
R5(config-if)#
R5(config-if)#no shut
R5(config-if)#
R5(config-if)#exit
R5(config)#
R5(config)#
*Oct  2 15:52:09.965: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R5(config)#

R5(config)#
R5(config)#interface eth0/1
R5(config-if)#ip add 129.70.1.1 255.255.255.192
R5(config-if)#
R5(config-if)#no shut
R5(config-if)#
R5(config-if)#exit
R5(config)#
R5(config)#
```

```
R6
Ethernet0/3          unassigned      YES TFTP    down      down

R6#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#
R6(config)#
R6(config)#interface eth0/0
R6(config-if)#ip add 129.70.1.70 255.255.255.252
R6(config-if)#no shut
R6(config-if)#
R6(config-if)#exit
R6(config)#
R6(config)#interface
*Oct  2 15:52:29.498: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R6(config)#interface eth0/1
R6(config-if)#
R6(config-if)#ip add 129.70.0.1 255.255.255.0
R6(config-if)#no shut
R6(config-if)#
R6(config-if)#exit
R6(config)#
R6(config)#
```

